

The Rejection Region Approach and the Z-test

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1. Overview

Key point: In the rejection region approach we compute a test statistic and reject H_0 when it falls in the tail region whose area is the significance level α .

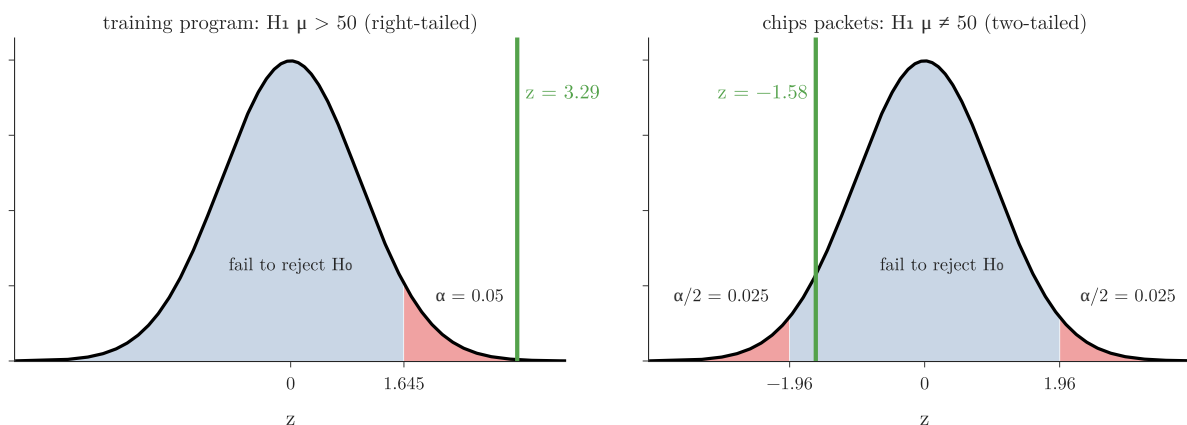


Figure 1: The two examples of this Note: the observed z (green) against the rejection region (red), $\alpha = 0.05$

The [null and alternative hypotheses Note](#) listed the eight steps of a hypothesis test. Here we run them end to end with the one-sample **z-test** on two problems. Figure 1 shows how each ends: the training program's z of 3.29 lands in the red rejection region, the chips' z of -1.58 does not.

This Note covers:

- the significance level α ;
- the z statistic and the assumptions of the one-sample z -test;
- the rejection region and the critical value;
- two complete examples, one right-tailed and one two-tailed;
- how α moves the rejection region;
- the weakness of the rejection region approach, which p -values fix.

2. Terms we build on

Key point: The standard normal curve and z -table, the central limit theorem with its standard error, and critical values.

- **Standard normal and z -table:** a z -score has the standard normal distribution, and the z -table gives the area to the left of any z (see the [standard normal Note](#)).
- **Central limit theorem and standard error:** for a sample of $n \geq 30$, the sample mean $\bar{X}$ is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$, the standard error (see the [central limit theorem Note](#)).

- **Critical value:** the z value that leaves a given area in the tail; 1.96 leaves 2.5% in each tail (see the [z-procedure Note](#)).

3. The significance level α

Key point: α is the probability of rejecting H_0 when it is actually true; we fix it before the test, usually at 0.05.

The **significance level**, written α , is a threshold fixed before the test. It decides whether H_0 will be rejected. In words:

α is the probability of rejecting the null hypothesis when it is actually true.

With $\alpha = 0.05$, if we ran many tests in which H_0 was in fact true, about 5 in 100 would still reject it. This mistake is called a Type I error (see the [errors, power and tails Note](#)).

The usual choices are 0.05 (5%) and 0.01 (1%). For most problems 0.05 works well and is treated as the standard. In a specialised field, a domain expert may choose another value from knowledge of the domain.

The significance level and the confidence level are related but not the same. A 95% confidence level corresponds to $\alpha = 0.05$: confidence level = $1 - \alpha$ (see the [z-procedure Note](#)).

We must fix α before the test. Without it there is no boundary between “reject” and “fail to reject”, and choosing it after seeing the result would let us pick whatever answer we like.

4. The one-sample z -test

Key point: The z -test measures how many standard errors the sample mean lies from the mean claimed by H_0 ; it needs σ and a normal sampling distribution.

4.1 Assumptions

The one-sample z -test can be used when:

1. **Normality:** the population is normal, or the sample has at least 30 values, so that by the central limit theorem $\bar{X}$ is approximately normal;
2. **Known σ :** the population standard deviation is known;
3. **Random sample:** the sample is drawn at random.

If σ is unknown, we use the t -test instead (see the [one-sample \$t\$ -test Note](#)).

4.2 The z statistic

If H_0 is true, the population mean is μ_0 , the value in H_0 . The sample mean then varies around μ_0 with standard error $\sigma/\sqrt{n}$, so standardizing it gives a standard normal variable.

1. **In words:** the distance between the sample mean and the mean claimed by H_0 , measured in standard errors.

2. Formula:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

3. **Example:** $\bar{x} = 53$, $\mu_0 = 50$, $\sigma = 5$, $n = 30$:

$$z = \frac{53 - 50}{5/\sqrt{30}} = \frac{3}{0.913} = 3.29$$

We divide by $\sigma/\sqrt{n}$, not by σ , because we are standardizing a sample **mean**, whose spread is the standard error.

5. The rejection region and the critical value

Key point: The rejection region is the tail of the standard normal curve with area α , on the side that H_1 points to; its boundary is the critical value.

The **rejection region** (also called the **critical region**) is the set of values of the test statistic for which we reject H_0 . It lies in the tail or tails that H_1 points to, and its area is α . The rest of the curve is the region where we fail to reject H_0 .

The **critical value** is the boundary of the rejection region:

- H_1 **with** $>$ (**right-tailed**): all of α in the right tail. For $\alpha = 0.05$ we need the z with area 0.95 to its left. The z -table has 0.9495 at 1.64 and 0.9505 at 1.65, so the critical value lies between them: 1.645.
- H_1 **with** $<$ (**left-tailed**): all of α in the left tail: -1.645 .
- H_1 **with** $\neq$ (**two-tailed**): $\alpha/2 = 0.025$ in each tail: ± 1.96 .

The decision rule is simple: if the test statistic falls in the rejection region, reject H_0 ; otherwise, fail to reject it.

Extra: One-sided and two-sided critical values are easy to mix up. At $\alpha = 0.05$, a one-tailed test uses 1.645 and a two-tailed test uses 1.96 (see the [t-procedure Note](#) for the same point with t). Using 1.96 for a one-tailed test makes it stricter than intended (a real α of 0.025); using 1.645 for a two-tailed test makes it looser (a real α of 0.10).

Python: Critical values come from `ppf`, the inverse of the CDF.

```
from scipy import stats
```

```
stats.norm.ppf(0.95)      # 1.645: right-tailed, alpha = 0.05
stats.norm.ppf(0.05)      # -1.645: left-tailed
stats.norm.ppf(0.975)     # 1.960: two-tailed, alpha = 0.05
```

6. Example: did a training program raise productivity?

Key point: $z = 3.29$ lies beyond the critical value 1.645, so we reject H_0 : the training program raised mean productivity.

A car factory makes **50 cars per day** on average, with a known population standard deviation of **5**. Some days it makes 45, some days 55. To raise output, the company runs a training program for its employees. Afterwards it measures the productivity of a random sample of **30** employees and finds a sample mean of **53 units per day**. Has the training significantly increased productivity?

1. **Hypotheses.** No benefit, against an increase:

$$H_0 : \mu = 50, \quad H_1 : \mu > 50$$

2. **Significance level.** $\alpha = 0.05$.
3. **Assumptions.** $n = 30$, so by the central limit theorem $\bar{X}$ is approximately normal. $\sigma = 5$ is known.
4. **Test.** Normal and σ known: the one-sample z-test.
5. **Test statistic.** z .
6. **Conduct the test.** As in section 4.2:

$$z = \frac{53 - 50}{5/\sqrt{30}} = 3.29$$

7. **Decide.** H_1 uses $>$, so the rejection region is the right tail beyond 1.645. Since $3.29 > 1.645$, z falls in the rejection region (Figure 1, left): we **reject** H_0 . The data gives strong evidence against H_0 , in favour of H_1 .
8. **Interpret.** The training program increased mean daily productivity above 50 units.

Python: The whole test in a few lines.

```
import numpy as np
from scipy import stats

z = (53 - 50) / (5 / np.sqrt(30))    # 3.29
z_crit = stats.norm.ppf(1 - 0.05)   # 1.645
z > z_crit                          # True: reject H0
```

7. Example: do chips packets weigh 50 g?

Key point: $z = -1.58$ lies between -1.96 and 1.96 , so we fail to reject H_0 : the sample does not show that the mean weight differs from 50 g.

A snack company claims that its packets of chips weigh **50 g** on average. A consumer watchdog wants to know whether the true mean weight differs from 50 g, more or less. It buys a random sample of **40** packets and finds a mean weight of **49 g**. The population standard deviation is known: **4 g**.

1. **Hypotheses.**

$$H_0 : \mu = 50, \quad H_1 : \mu \neq 50$$

2. **Significance level.** $\alpha = 0.05$.

3. **Assumptions.** $n = 40 \geq 30$: $\bar{X}$ is approximately normal. $\sigma = 4$ is known.

4. **Test.** One-sample z-test.

5. **Test statistic.** z .

6. **Conduct the test.**

$$z = \frac{49 - 50}{4/\sqrt{40}} = \frac{-1}{0.632} = -1.58$$

7. **Decide.** H_1 uses $\neq$: we do not know the direction, so the test is two-tailed. $\alpha = 0.05$ is split into 0.025 in each tail, giving critical values ± 1.96 . Since $-1.96 < -1.58 < 1.96$, z falls between them (Figure 1, right): we **fail to reject** H_0 .

8. **Interpret.** The 40 packets do not give enough evidence that the mean weight differs from 50 g. The watchdog has no case against the company.

This does not prove that the packets weigh exactly 50 g on average. As the [null and alternative hypotheses Note](#) showed, failing to reject H_0 only means the evidence was not strong enough.

8. How α moves the rejection region

Key point: A smaller α pushes the critical values outward and shrinks the rejection region, so rejecting a true H_0 becomes rarer.

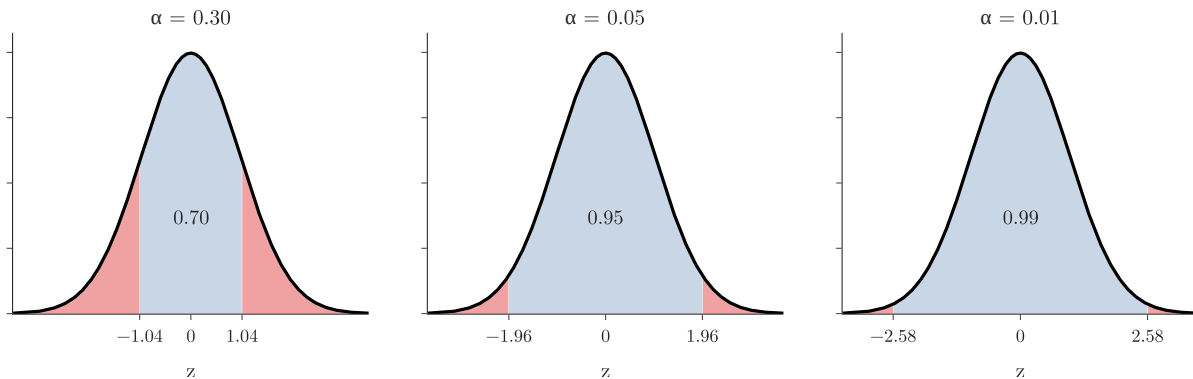


Figure 2: Two-tailed rejection regions (red) for three significance levels; the middle area is $1 - \alpha$

Figure 2 shows a two-tailed test at three significance levels:

α	Area in each tail	Critical values
0.30	0.15	± 1.04
0.05	0.025	± 1.96

α	Area in each tail	Critical values
0.01	0.005	± 2.58

- **Lower** α (5% to 1%): the critical values move out and the region where we fail to reject grows. A true H_0 is rejected less often.
- **Higher** α (say 30%): the rejection region grows. Even when H_0 is true, the statistic now often lands in the rejection region by chance, and we wrongly reject H_0 .

This is the meaning of “the probability of rejecting H_0 when it is actually true”: it is exactly the red area. A lower α has a cost too, which the [errors, power and tails Note](#) explains.

9. The weakness of the rejection region approach

Key point: The rejection region approach only says on which side of a boundary the statistic fell; it cannot say how strong the evidence is.

The rejection region approach gives a yes-or-no answer, and that causes two problems.

A knife-edge boundary. In a two-tailed test at $\alpha = 0.05$, $z = 1.97$ means “reject” and $z = 1.95$ means “fail to reject”. A difference of 0.02 flips the decision, although the two samples are almost equally convincing.

No strength of evidence. $z = 2$ and $z = 15$ both fall in the rejection region and both lead to “reject H_0 ”. But they are very different:

z	Probability of a value at least this large if H_0 is true	Evidence against H_0
2	0.023	moderate
15	about 4×10^{-51}	overwhelming

A z of 15 is so unlikely under H_0 that it leaves almost no doubt, while a z of 2 is the kind of value that turns up 2 or 3 times in 100 by chance. The rejection region approach treats them the same.

The fix is to compute one more number, the **p-value**, which measures the strength of the evidence against H_0 . It is the approach used in practice, and the subject of the [p-values Note](#).

10. Summary

	Training program	Chips packets
H_0 / H_1	$\mu = 50 / \mu > 50$	$\mu = 50 / \mu \neq 50$
Data	$n = 30, \bar{x} = 53, \sigma = 5$	$n = 40, \bar{x} = 49, \sigma = 4$
Tails	right	two
Critical value ($\alpha = 0.05$)	1.645	± 1.96
z	3.29	-1.58
Decision	reject H_0	fail to reject H_0

- α , fixed in advance, is the probability of rejecting a true H_0 ; 0.05 is the standard choice.
- The one-sample z-test needs normality (or $n \geq 30$), a known σ and a random sample; $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.
- The rejection region has area α in the tail(s) H_1 points to; its boundary is the critical value.
- Right-tailed 1.645, two-tailed ± 1.96 at $\alpha = 0.05$.
- A smaller α shrinks the rejection region.
- The approach cannot tell weak from overwhelming evidence; p-values can.

11. Key terms

Term	Meaning
Significance level (α)	The probability of rejecting H_0 when it is actually true; fixed before the test, usually 0.05
Z-test (one-sample)	A test of a population mean when σ is known and $\bar{X}$ is normal, using $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
Z statistic	The value of z computed from the sample in a z-test
Rejection region (critical region)	The values of the test statistic for which we reject H_0 ; its area under H_0 is α
Critical value	The boundary of the rejection region, e.g. 1.645 (right-tailed) or ± 1.96 (two-tailed) at $\alpha = 0.05$
Strength of evidence	How strongly the data speaks against H_0 ; the rejection region approach does not measure it